

Apex Systems

Secure, scalable enterprise architecture for a controlled procurement transformation

Artificial / Synthetic Supplier Proposal

Procurement Automation & Supplier Evaluation Platform

Executive Summary

Apex Systems proposes a security-led procurement platform designed for organizations that need strong integration, auditability and controlled growth. The proposed architecture uses an API-first application layer, a relational transaction store, document processing services and configurable workflow controls. Apex emphasizes technical depth and security governance, while acknowledging a higher total implementation price and a moderate delivery schedule.

Understanding of the Requirement

- Create a single workspace for procurement requests, supplier submissions, evaluation criteria, evidence and decisions.
- Reduce manual comparison by extracting proposal content and presenting criterion-level evidence to evaluators.
- Maintain an auditable record of scoring inputs, approvals, changes and final supplier decisions.
- Integrate with identity, document storage and existing enterprise systems without forcing a replacement of core systems.

Proposed Solution

- API gateway and web application layer with role-based access control.
- Document ingestion service supporting PDF and office-document conversion, text extraction and metadata capture.
- Evaluation service that sends structured criteria and extracted proposal content to an LLM, with deterministic validation before scoring.
- PostgreSQL transaction database with encrypted backups and immutable audit-event records.
- Integration adapters for SSO, enterprise content storage and notification services.

Proposal positioning: This document is intentionally synthetic and designed for classroom evaluation. It contains explicit strengths, limitations, assumptions and evidence so that an evidence-grounded LLM evaluator can distinguish suppliers rather than assigning identical generic scores.

Solution Architecture & Implementation

The following architecture is presented at proposal level. Where a supplier has not provided enough detail, the document explicitly identifies the item as an assumption or discovery dependency.

Layer	Proposed approach	Evidence / assumption
Experience	Responsive web portal with buyer, evaluator and administrator roles.	Role matrix and approval workflow included.
Application	Containerized services behind an API gateway.	Designed for horizontal scaling; exact peak volumes to be confirmed.
Data	Managed relational database plus object storage for source proposals.	Encryption at rest and backup retention are included.
AI	Structured LLM extraction/evaluation with schema validation and prompt versioning.	Human review remains available for exceptions.
Integration	REST APIs, SSO via OIDC/SAML, outbound webhooks.	Legacy connector effort is an implementation assumption.

Implementation Timeline

Phase	Duration	Key milestones
Discovery & design	3 weeks	Workshops, process maps, security design, backlog approval
Build foundation	5 weeks	Core portal, identity, data model and audit logging
Evaluation & integrations	5 weeks	Document pipeline, evaluation workflow, priority integrations
Testing & pilot	3 weeks	Security testing, UAT, performance checks, pilot
Production rollout	2 weeks	Training, cutover and hypercare

Delivery Team & Governance

Role	Allocation	Responsibility
Program Manager	0.5 FTE	Governance, plan, risks and stakeholder reporting
Solution Architect	0.5 FTE	Architecture, integration and technical decisions
Backend Engineers	2 FTE	APIs, workflows, persistence and integrations
AI/Data Engineer	1 FTE	Document pipeline, structured LLM evaluation and validation
Security Engineer	0.5 FTE	Threat modeling, controls and testing
QA Engineer	1 FTE	Functional, regression and performance testing

Delivery governance: The supplier will provide weekly delivery reporting covering completed work, next milestones, decisions, risks and dependencies. Customer sign-off is expected at agreed phase gates.

Commercial Proposal

The following figures are fictional and intended only to create meaningful commercial differences among the four artificial suppliers. Pricing assumptions are part of the evidence available to the evaluator.

Component	One-time	Annual / recurring
Discovery & solution design	\$42,000	—
Core platform implementation	\$168,000	—
AI evaluation workflow	\$76,000	—
Security hardening & testing	\$38,000	—
Integration allowance	\$54,000	—
Training & rollout	\$22,000	—
Platform support	—	\$72,000/year
Estimated total	\$400,000	\$72,000/year

Commercial Assumptions

- Pricing is intentionally positioned above the other synthetic suppliers because the proposal includes a deeper security and integration work package.
- Cloud hosting, third-party LLM consumption and unusual legacy-system adapters are excluded unless specifically contracted.
- Estimate assumes up to 20 business integrations, 500 active suppliers per procurement event and 2,000 proposal documents per month.

Security, Compliance & Risk Controls

- TLS 1.2+ for network communication and AES-256-class encryption for stored data.
- Role-based access control with least-privilege administration and separate evaluator permissions.
- SSO integration and optional MFA enforcement through the customer's identity provider.
- Central audit trail covering authentication, document access, scoring changes, approvals and exports.
- Quarterly vulnerability scanning and annual penetration testing are included in the support plan.
- Proposal data is not used to train a shared public model; model/provider configuration remains subject to customer approval.

Risk Register

Risk	Mitigation	Residual concern
Legacy integration complexity	Prototype priority interfaces during discovery.	Exact effort depends on undocumented interfaces.
LLM extraction errors	Schema validation, confidence flags and human review for exceptions.	AI output remains probabilistic.
Change requests	Baseline scope and use a controlled change process.	Additional integrations may affect schedule.
Security approval delay	Start threat modeling in week 1.	Customer review timelines are outside supplier control.

Support, Experience & References

- 24x7 incident intake for severity-1 production issues; business-hours support for standard requests.
- Named service manager and monthly service review.
- Target response: 30 minutes for severity 1, 2 hours for severity 2, next business day for severity 3.
- Quarterly platform health review covering availability, security findings, usage and backlog.
- Apex states experience across enterprise workflow, document automation and security-sensitive systems; specific customer references can be provided under the procurement process.

Relevant Experience / References

- Reference A — Synthetic enterprise procurement modernization, 14-month delivery, multi-system integration.
- Reference B — Synthetic regulated document workflow, 9-month implementation, audit-heavy operating model.

Known Limitations & Items Requiring Clarification

- Higher commercial price than the other synthetic proposals.
- Moderate delivery schedule rather than the fastest launch.
- Legacy integration effort and peak-volume assumptions require discovery confirmation.
- Specific named customer references are not supplied in this artificial document.

Traceability Notes for Evaluation

This proposal is written so that each evaluation criterion can be assessed from document evidence: **Technical Capability** from architecture and integration sections; **Implementation Plan** from timeline, milestones and team; **Commercial Value** from the price table and assumptions; **Security & Compliance** from controls, limitations and risk register; and **Support & Experience** from the service model and references.

Important: Any statement not explicitly supported in this proposal should be treated as missing evidence by an evaluator. The supplier should not receive credit merely because a capability would be reasonable or common in the industry.